



TECHNICAL UNIVERSITY

OF CLUJ-NAPOCA, ROMANIA

FACULTY OF AUTOMATION AND COMPUTER SCIENCE

THE TITLE OF THE THESIS

LICENSE THESIS

Graduate: **Firstname LASTNAME**

Supervisor: **scientific title Firstname LASTNAME**

2026



TECHNICAL UNIVERSITY

OF CLUJ-NAPOCA, ROMANIA

FACULTY OF AUTOMATION AND COMPUTER SCIENCE

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Graduate: **Firstname LASTNAME**

THE TITLE OF THE THESIS

1. **Project proposal:** *Short description of the license thesis and initial data*
2. **Project contents:** *(enumerarea părților componente) Example: : (enumerate the main component parts) Presentation page, advisor's evaluation, title of chapter 1, title of chapter 2, ..., title of chapter n, bibliography, appendices .*
3. **Place of documentation:** *Example:* Technical University of Cluj-Napoca, Computer Science Department
4. **Consultants:**
5. **Data of issue of the proposal:** October 1, 2025
6. **Data of delivery :** July 10, 2026

Graduate: _____

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elaborată în vederea susținerii examenului de finalizare a studiilor de licență la Facultatea de Automatică și Calculatoare, Programul de studiu _____ din cadrul Universității Tehnice din Cluj-Napoca, sesiunea _____ a anului universitar _____, declar pe proprie răspundere, că această lucrare este rezultatul propriei activități intelectuale, pe baza cercetărilor mele și pe baza informațiilor obținute din surse care au fost citate, în textul lucrării și în bibliografie.

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Data

Nume, Prenume

Semnătura

General Advice

READ FIRST (this page should be removed from the final version):

1. If you print the thesis on paper, the three preceding pages (title page, summary page and declaration) should be printed on separate pages (one-side) and should be included in the printed paper. The summary page (the second one) must be signed by the graduate and the supervisor. The date on the declaration should be the date when the thesis is handed to the commissions' secretary.
2. On the title page, you should include the correct scientific title of your supervisor (use the Department web pages to find that).
3. Every chapter begins on a new page.
4. Do not change page margins.
5. Obey the other instructions provided in every chapter.
6. This template obeys all the formatting rules.
7. We included the `hyperref` package to generate navigation links. To produce a version for printing on paper uncomment the line containing `%\hypersetup{hidelinks}` located in the beginning of the main file `thesis_eng.tex`.

Contents

Chapter 1	Introduction - Project Context	1
Chapter 2	Project Objectives	2
Chapter 3	Bibliographic Research	3
3.1	A Section Name	4
3.2	Another Section Name	4
3.2.1	A Subsection Name	4
Chapter 4	Analysis and Theoretical Foundation	5
4.1	A section title	5
4.1.1	A Subsection title	5
Chapter 5	Detailed Design and Implementation	6
Chapter 6	Testing and Validation	7
Chapter 7	User's Manual	8
Chapter 8	Conclusions	9
	Bibliography	10
	Appendix A Relevant Code sections	11
	Appendix B Other Relevant Info	12
	Appendix C Published Papers	13

Chapter 1. Introduction - Project Context

Should take 2 to three pages of the paper.

What should be here:

- framing the design theme in the contemporary context
- an outline of the exact domain of the design theme

Chapter 2. Project Objectives

Should take 2 to 3 pages.

Here you describe the design/research theme, precisely formulated, with clear objectives on 2-3 pages and, possibly, explanatory figures.

Chapter 3. Bibliographic Research

This chapter should take between 3 and 10 pages.

Bibliographic research has as an objective the establishment of the references for the project, within the project domain/thematic. While writing this chapter (in general the whole document), the author will consider the knowledge accumulated from several dedicated disciplines in the second semester, 4th year (Project Elaboration Methodology, etc.), and other disciplines that are relevant to the project theme.

Each reference **must** be cited within the document. Please look at the examples below (depending on the project theme, the presentation of a method/application can vary).

Referințele are incluse în capitolul Bibliography.

References can be managed with JabRef, an application which can be downloaded from <https://www.jabref.org/#download>

Examples of what should be included in each type of reference can be found at here.

About common errors found in online libraries of references you can read at here

In Chapter 4 of [?], Spitzner discusses the advantages and disadvantages of honeypot systems.

References will be included in the Bibliography section. The reference format must be IEEE, or similar. The introduction of new references in the Bibliography section, and their citation within the document text can be done manually (by obeying the format), but it is not recommended as it is not easy to manage them, or by using the tools mentioned in the last paragraphs of this chapter.

In the Bibliography section, there are examples of references to conferences or workshops articles [?], journal [?], and books [?], [?]. References to applications or online resources (web pages) must include at least a short relevant description in addition to the link [?], and other information is available (authors, year, etc.). References that contain only the link to the online resource will be placed in the page footer.

Each reference must be cited within the document text, see example below (depending on the project theme, the presentation of a method/application can vary).

In paper [?] the authors present a system for moving obstacle detection using stereo-vision and an estimation of own movement.

The method is based on ... autorii prezintă un sistem pentru detecția obstacolelor în mișcare folosind stereo viziune și estimarea mișcării proprii. Metoda se bazează pe ...*discuss the algorithms, data structures, functionality, specific aspects related to the project theme, etc.* ... Discussion: *pros and cons*.

3.1. A Section Name

3.2. Another Section Name

3.2.1. A Subsection Name

DO NOT copy technology descriptions here

Chapter 4. Analysis and Theoretical Foundation

Together with the next two chapters takes about 70% of the whole paper.

The purpose of this chapter is to explain the operating principles of the implemented application. Here you write about your solution from a theory standpoint – i.e. you explain it and demonstrate its theoretical properties/value, e.g.:

- used or proposed algorithms,
- used protocols,
- abstract models,
- logic explanations/arguments concerning the chosen solution,
- logic and functional structure of the application, etc.

YOU SHOULD NOT write about the implementation. YOU SHOULD NOT describe technologies and other things which do not pertain to your project (no fillers, please!).

4.1. A section title

4.1.1. A Subsection title

Every table in the thesis should be numbered as Table $x.y$, where x is the chapter number where the table is included, and y is the number of the table within that chapter. There should be one empty line after the paragraph preceding the table, and one empty line after the table. Example: in this row we have inserted a reference to Table 4.1.

Table 4.1: Results

Case	Method#1	Method#2	Method#3
1	50	837	970
2	47	877	230
3	31	25	415

Every figure used in the thesis should be referred (e.g. Figure $x.y$ shows the components of the system...) and numbered.

Numbering is like this: Figure $x.y$ where x is the chapter number and y is the number of the figure within that chapter. E.g. , in this row we have inserted a reference to Figure 4.1.

Figure 4.1: Figure name

Chapter 5. Detailed Design and Implementation

Together with the previous and next chapter takes about 70% of the paper.

The purpose of this chapter is to document the developed application such a way that it can be maintained and developed later. A reader should be able (from what you have written here) to identify the main functions of the application.

The chapter should contain (but not limited to):

- a general application sketch/scheme,
- a description of every component implemented, at module level,
- class diagrams, important classes and methods from key classes.

Chapter 6. Testing and Validation

Together with the previous two chapters should take about 70% of the paper.

Chapter 7. User's Manual

In the section describing the installation procedure you should detail the hardware and software resources needed for installing and running the application, and a step by step description of how your application can be deployed/installed. An administrator should be able to perform the installation/deployment based on your instructions.

In the section for the user you should describe how to use the application from the point of view of a user with no inside technical information; this should be adorned with screen shots and a stepwise explanation of the interaction. Based on user's manual, a person should be able to install and use your product.

Should take 1 to 5 pages.

Chapter 8. Conclusions

This chapter should take one or two pages.

In this chapter you should include::

- A summary of your contributions/achievements,
- A critical analysis of the results achieved,
- A description of the possibilities of improvements/further development.

Bibliography

Appendix A. Relevant Code sections

```
/** Maps are easy to use in Scala. */
object Maps {
  val colors = Map("red" -> 0xFF0000,
                   "turquoise" -> 0x00FFFF,
                   "black" -> 0x000000,
                   "orange" -> 0xFF8040,
                   "brown" -> 0x804000)

  def main(args: Array[String]) {
    for (name <- args) println(
      colors.get(name) match {
        case Some(code) =>
          name + " has code: " + code
        case None =>
          "Unknown color: " + name
      }
    )
  }
}
```


Appendix B. Other Relevant Info

Proofs etc. if any. Otherwise remove this chapter

Appendix C. Published Papers

If any. Otherwise remove this chapter